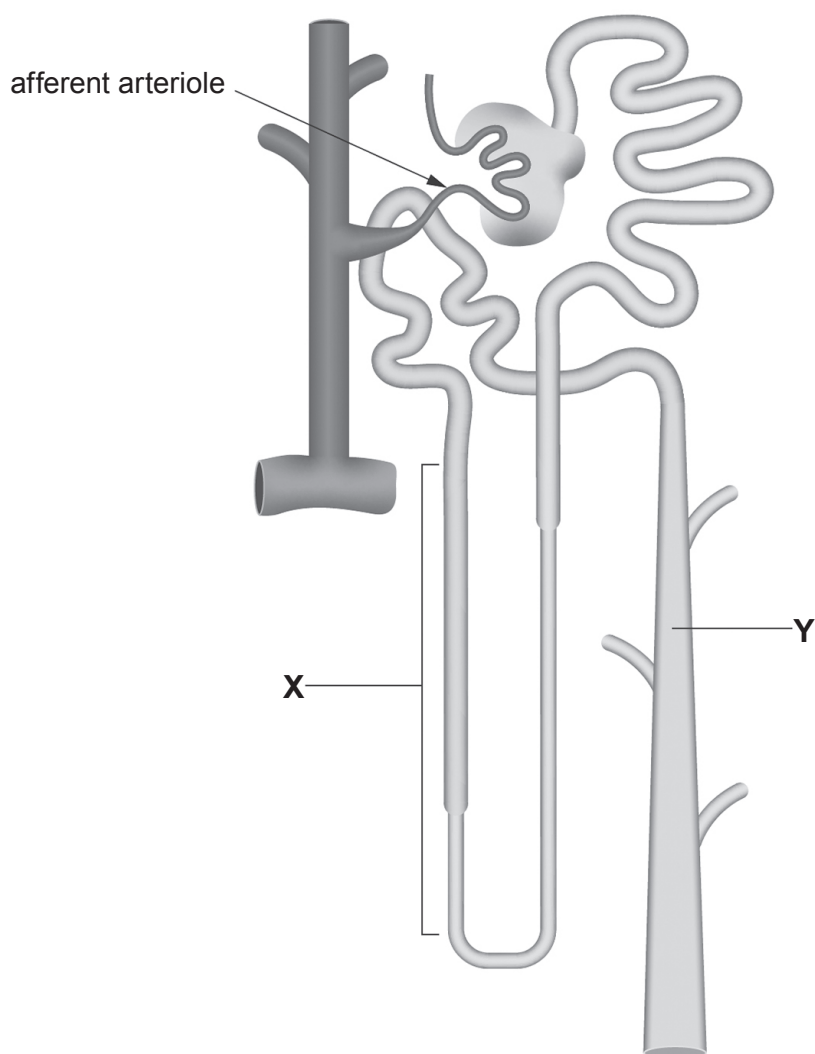


5. The mammalian kidney has a role in two physiological processes, excretion and homeostasis. **Image 5.1** shows a single kidney nephron. The strategy that the kidney uses for excretion is ultrafiltration followed by selective reabsorption.

Image 5.1



- (a) (i) Use **labelled lines** on **Image 5.1** to show the sites of:
- I. ultrafiltration
 - II. selective reabsorption

[2]



(ii) Structures labelled **X** and **Y** in **Image 5.1** are involved in homeostasis. Name structures **X** and **Y** and the homeostatic process in which they are involved. [2]

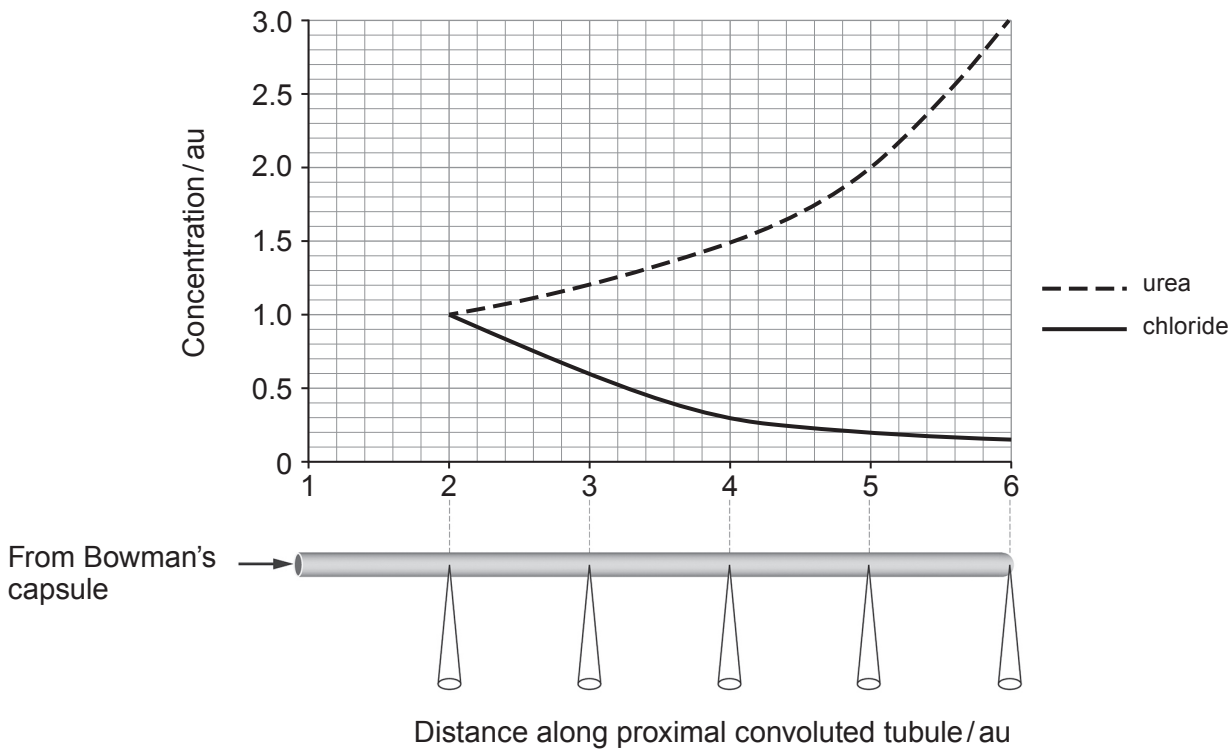
X

Y

Homeostatic process

The use of micropipettes has allowed samples of fluid to be withdrawn from specific points along kidney tubules of experimental animals. Samples of filtrate were taken from five positions along the proximal convoluted tubule and the concentrations of urea and chloride ions were measured. The results are shown in **Graph 5.2**.

Graph 5.2



(b) (i) Use **Graph 5.2** to explain the change in concentrations for urea and chloride ions along the tubule. [2]

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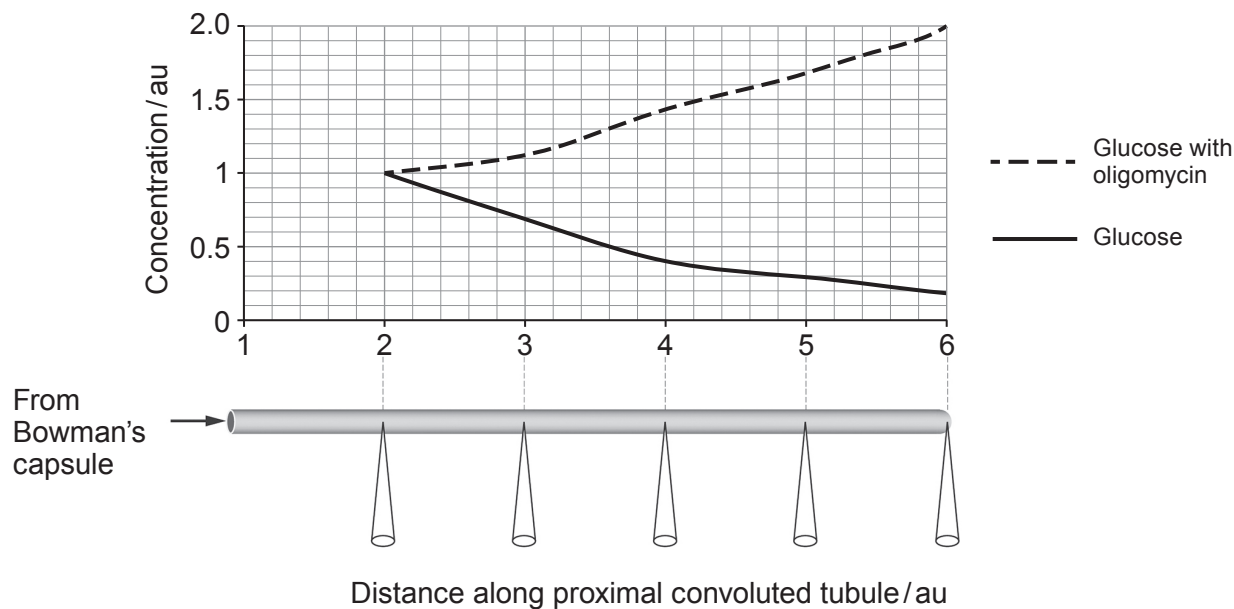
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The experiment was repeated but concentrations of glucose were measured with and without oligomycin.

Oligomycin is a chemical compound that specifically inhibits respiration. The results are shown in **Graph 5.3**.

Graph 5.3



- (ii) Use **Graph 5.3**, to explain the change in concentration along the tubule for glucose with **and** without the respiratory inhibitor oligomycin.

[4]

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